Lab 7: Fuzzing for Input Validation Bugs

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Fuzz testing of parse_applicant, marks_average and payable_award using Hypothesis. Steps 2, 4, 5 and 6 of the lab handout are shown in order.

Step 2: Run the buggy application

Original scholarship_awards.py run with sample inputs (merit:82, 72,85,90, 5000|500). It works on well-formed input; the bugs only appear on unexpected input.

```
$ python scholarship_awards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

Step 4: Hypothesis tests failing on the buggy code

All 3 fuzz tests fail. Hypothesis found crashing inputs such as None, '' (empty string), ':' and '|' (AttributeError / ValueError). The repeated pytest internal traceback in the middle is shortened in the image; the failing cases and the summary are shown.



```
$ python -m pytest test_scholarship_awards.py --tb=short -p no:cacheprovider
============================= test session starts ==============================
platform darwin -- Python 3.13.5, pytest-8.3.4, pluggy-1.5.0
rootdir: /private/tmp/lab7_before
plugins: hypothesis-6.170.15, anyio-4.16.0
collected 3 items

test_scholarship_awards.py FFF                                           [100%]

=================================== FAILURES ===================================
_____________________ test_parse_applicant_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ~~~~^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ~~~~~~~~~~~~~~~~~^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/logging.py", line 829, in _runtest_for
  |     yield
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/pluggy/_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "/opt/anaconda3/lib/python3.13/site-packages/_pytest/skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
... [337 lines of repeated traceback omitted] ...
    |     assert False, f"payable_award raised {e!r} for {s!r}"
    | AssertionError: payable_award raised AttributeError("'NoneType' object has no attribute 'split'") for None
    | assert False
    | Failing test case: test_payable_award_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_scholarship_awards.py::test_parse_applicant_no_crash - ExceptionG...
FAILED test_scholarship_awards.py::test_marks_average_no_crash - ExceptionGro...
FAILED test_scholarship_awards.py::test_payable_award_no_crash - ExceptionGro...
============================== 3 failed in 1.18s ===============================
```

Step 5: Run the fixed application

After hardening the functions (type checks, split-length checks, try/except around int() and float(), rejecting nan/inf, safe return values). Same results on valid input.

```
$ python scholarship_awards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

Step 6: Hypothesis tests passing on the fixed code

All 3 fuzz tests pass against the fixed scholarship_awards.py.

```
$ python -m pytest test_scholarship_awards.py -p no:cacheprovider
============================= test session starts ==============================
platform darwin -- Python 3.13.5, pytest-8.3.4, pluggy-1.5.0
rootdir: /Users/harish/L6_L7_44_scholarship_awards/Lab 7
plugins: hypothesis-6.170.15, anyio-4.16.0
collected 3 items

test_scholarship_awards.py ...                                           [100%]

============================== 3 passed in 0.18s ===============================
```